

SUBJECT	Lower Secondary Maths Year 8
TOPIC	Module Four
ASSIGNMENT	Four
STUDENT NAME	

FEEDBACK AND REFLECTION:

- ☐ I have read the feedback on my last assignment carefully.
- ☐ I have considered whether any of that feedback is relevant to this assignment.

One thing I will try to do either the same or differently in this assignment as a result of my previous feedback is: _____

ASSIGNMENT GUIDANCE

- Make sure you write your full name clearly in the box at the top of this page.
- Please **write** your answers **by hand** in **black ink** on this question paper, unless you have obtained specific permission to submit typed work.
- This is not a timed assignment. We estimate that this assignment may take you between 30 and 40 minutes. However, if you need to take longer, then please do so.
- You are not expected to complete your assignments without referring to your notes or coursebook.
- In order for maximum marks to be awarded, for any question worth more than one mark, please show your workings to make your method clear.
- You may use a calculator for Questions 5-9 in this assignment (unless otherwise stated).





LS Maths Assignment Four

Note. Questions 1-4 are non-calculator questions.

1.

a. Write each pair of fractions with the same denominator.

(3 marks)

i. $\frac{2}{3}$ and $\frac{5}{9}$

ii. $\frac{3}{4}$ and $\frac{4}{5}$

iii. $\frac{5}{8}$ and $\frac{7}{12}$

b. Write a suitable inequality sign between each pair of fractions.

(3 marks)

For example, for i. does this mean $\frac{2}{3} < \frac{5}{9}$, $\frac{2}{3} \leq \frac{5}{9}$, $\frac{2}{3} > \frac{5}{9}$ or $\frac{2}{3} \geq \frac{5}{9}$?

i. $\frac{2}{3}$ $\frac{5}{9}$

ii. $\frac{3}{4}$ $\frac{4}{5}$

iii. $\frac{5}{8}$ $\frac{7}{12}$

(Total 6 marks)





2. Cancel these fractions fully, then multiply. Give the simplest fraction for final answer. Please note: 1 mark will be awarded for cancelling the fractions before you multiply, so as always, be sure to make your method clear.

Work out:

a. $\frac{6}{9} \times \frac{3}{4} \times \frac{12}{36}$

(2 marks)

b. $\frac{4}{10} \times \frac{7}{12} \times \frac{20}{14}$

(2 marks)

c. $\frac{11}{16} \times \frac{12}{24} \times \frac{18}{22}$

(2 marks)

(Total 6 marks)





3. Work out (remember BIDMAS). Full workings must be seen. **Note: remember to use your cancelling fractions skill.**

a. $\frac{200}{300} \div \frac{40}{90} + \frac{3}{4}$

(2 marks)

b. $\frac{2}{9} + \frac{5}{15} \div \frac{6}{30}$

(2 marks)

c. $\frac{4}{6} \times \frac{4}{5} - \frac{7}{15}$

(2 marks)

d. $\frac{10}{12} - \frac{6}{8} \times \frac{10}{11}$

(2 marks)

(Total 8 marks)





4. At a rowing regatta $\frac{1}{3}$ of the medals won by the Paddle Power Rowing club were gold, and $\frac{3}{8}$ were silver. What fraction were bronze?

(2 marks)

Note. Calculators may be used for the remainder of this assignment (unless otherwise stated).

5. Anna plays rugby and has been practising taking conversions.
Last week she scored 24 out of 40.
This week she scored 19 out of 30.
Has her scoring rate improved? Explain your answer, with working shown.

(3 marks)



6. A couple pays £1200 for a holiday in Portugal. Unfortunately, due to the pandemic the holiday company cancels the holiday and gives the customers this option:

Rebook another holiday and get a 20% DISCOUNT incentive*

We have a range of alternative holidays available for you to choose from and would love to take you away on holiday. Plus, if the holiday you book is more expensive than your original holiday, **you'll automatically receive an incentive worth up to 20% of the value of the holiday***.

- a. The customers find an alternative holiday in Greece costing £1800, for them both, before discounts.
Use a **non-calculator** method to show what the cost of this holiday will be once the discount mentioned above has been applied.

(3 marks)

- b. Using online discount code SUMMER gives a further £100 off the cost of the holiday, so what is the final total holiday price?

(1 mark)

- c. Given that the customers had already paid the holiday company £1200 for the holiday in Portugal, and are paying an equal share towards this holiday, how much extra will it cost them **each**?

(2 marks)

(Total 6 marks)



7. The same bike is for sale in two shops. In Shop A its original price is £198 and the offer is $\frac{1}{3}$ off.
In Shop B its original price is £239 and the offer is 40% off.
Work out the exact sale price of the bike in each shop.

Which shop offers the lower price?

(Total 3 marks)

8. Calculate the following using a suitable method:
- a. A new car depreciated (lost value) by 25% after one year. If it was bought for £12,000, show a **non-calculator method** you could use to find out how much it was worth after the first year?

(3 marks)





- b. A Wolsey Hall student increased in height from 1.52m to 1.68m in one year.

Calculate this percentage increase in height.

(3 marks)

(Total 6 marks)

TOTAL FOR ASSIGNMENT 40 MARKS

